

Activity 1: Descriptive Statistics

After data retrieval and preparation comes the exploration phase:

- Please give a high-level explanation of *data exploration*, and what this step aims to do.

Specifically, to achieve this, you need to explore the descriptive statistics of your data:

- What kind of features do you have in a dataset? Nominal, ordinal or numerical?
- List the statistics you would like to explore for each kind of feature.
- Why are these statistics important?
- What can you explore by examining these statistics?
- What if there is an Index column in the dataset?
- How can you explore these descriptive statistics in iPython (with pandas)?

Activity 2: Visualisation

Information becomes much easier to grasp when shown in a picture, therefore graphical techniques are widely used to gain an understanding of each feature, as well as the interactions between features. The following are a list of graphical representations:

- Pie Chart
- Bar Chart
- Line Graph
- Histogram Graph
- Distribution Graph
- Scatter Plot
- Boxplot

Please discuss:

- What kind of feature can be visualised by each kind of graph?
- How would you draw the corresponding graph in iPython?

(Hint: <https://www.youtube.com/watch?v=hZxnzfmt5v8#t=03m15s>)

Practical Exercise 1: Data Retrieving

Today, we will explore the abalone dataset. Please follow Practical exercise 2 in [Tute/lab 02](#) to retrieve the data.

Practical Exercise 2: Manipulating the Data

There are a variety of ways to access the value of a cell in a DataFrame. For example, let's access the 4th line of the 'Sex' column of the abalone data.

1. Access by specifying the **column** first, then the line of the **row**:

```
In [1]: abalone['Sex'][3]
Out[1]: 'M'
```

2. Access by using **.loc** attribute

```
In [2]: abalone.loc[3]['Sex']
Out[2]: 'M'
```

3. Access by using **.iloc** attribute

```
In [3]: abalone.iloc[3]['Sex']
Out[3]: 'M'
```

```
In [4]: abalone.iloc[3][0]
Out[4]: 'M'
```

Now, let's access a limited section of rows by using masks. E.g check the diameter information of those abalone whose rings are 21.

```
In [5]: mask_ring21 = abalone['Rings'] == 21
In [6]: abalone.loc[mask_ring21, 'Diameter']
Out[6]:
83      0.475
181     0.510
310     0.485
428     0.455
469     0.555
478     0.590
628     0.415
673     0.390
811     0.365
2180    0.420
2199    0.550
2344    0.550
```

```
3151    0.500
3930    0.535
Name: Diameter, dtype: float64
```

Check mean value by grouping a feature, e.g. check the mean value when grouping *abalone* by 'Rings':

```
In [7]: numeric_df = abalone.select_dtypes('number')
        numeric_df.groupby('Rings').mean()
Out[7]:
```

	Length	Diameter	Height	Whole weight	Shucked weight	\
Rings						
1	0.075000	0.055000	0.010000	0.002000	0.001000	
2	0.150000	0.100000	0.025000	0.015000	0.004500	
...						
29	0.700000	0.585000	0.185000	1.807500	0.705500	

	Viscera weight	Shell weight
Rings		
1	0.000500	0.001500
2	0.004000	0.005000
...		
29	0.321500	0.475000

Sort the abalone data by 'Length':

```
In [8]: abalone.sort_values(by='Length').head()
Out[8]:
```

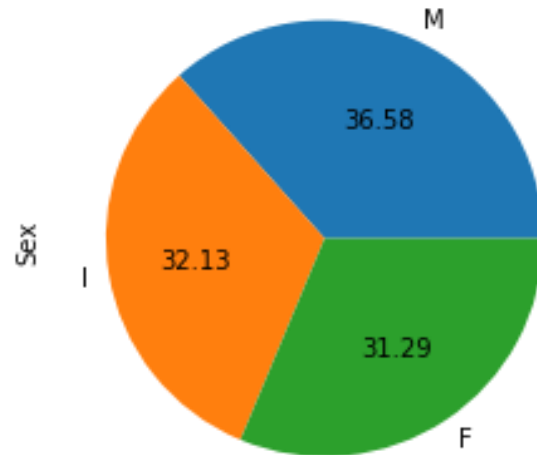
	Sex	Length	Diameter	Height	Whole weight	Shucked weight	\
236	I	0.075	0.055	0.010	0.0020	0.0010	
238	I	0.110	0.090	0.030	0.0080	0.0025	
2114	I	0.130	0.095	0.035	0.0105	0.0050	
237	I	0.130	0.100	0.030	0.0130	0.0045	
1986	I	0.135	0.130	0.040	0.0290	0.0125	

	Viscera weight	Shell weight	Rings
236	0.0005	0.0015	1
238	0.0020	0.0030	3
2114	0.0065	0.0035	4
237	0.0030	0.0040	3
1986	0.0065	0.0080	4

Practical Exercise 3: Visualisation

Draw the pie chart for 'Sex' column

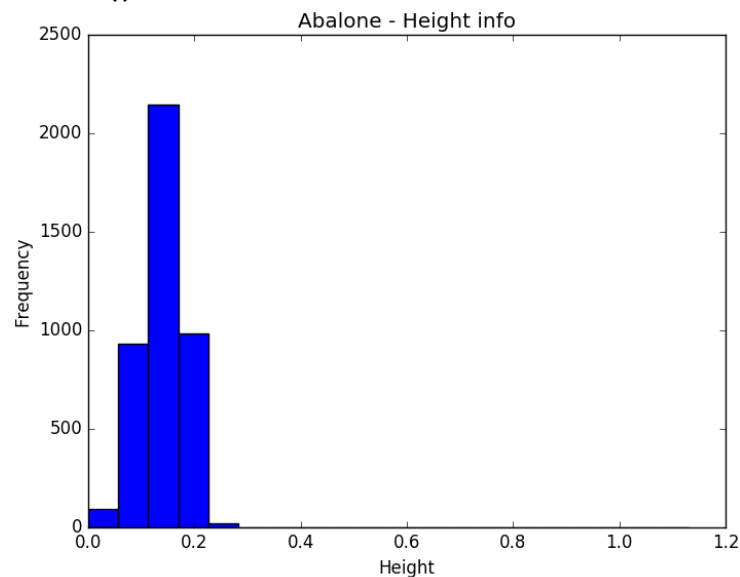
```
In [9]: import matplotlib.pyplot as plt
In [10]: abalone['Sex'].value_counts().plot(kind='pie', autopct='%0.2f')
         plt.show()
```



Note: In the most recent versions of Jupyter Notebook, a graph is displayed automatically without having to call the `plt.show()` function. However, you have to call this function explicitly if you are to write a python script.

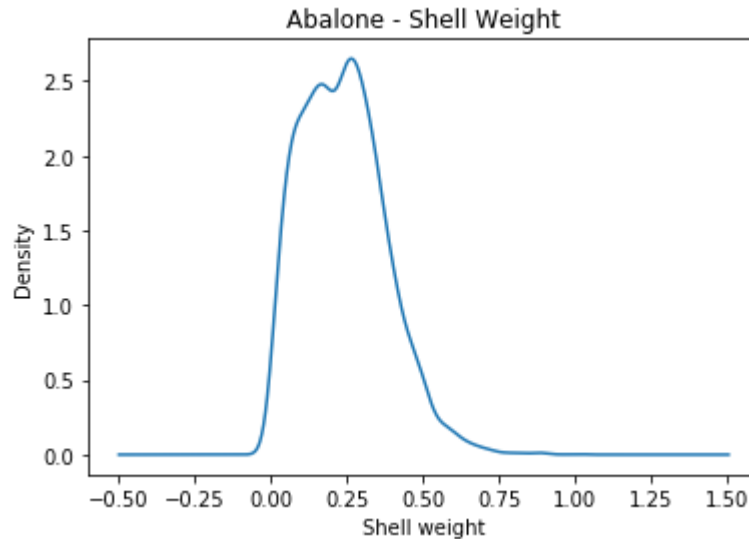
Draw the histogram for 'Height':

```
In [11]: abalone['Height'].plot(kind='hist',bins=20)
plt.title('Abalone - Height info')
plt.xlabel('Height')
plt.show()
```



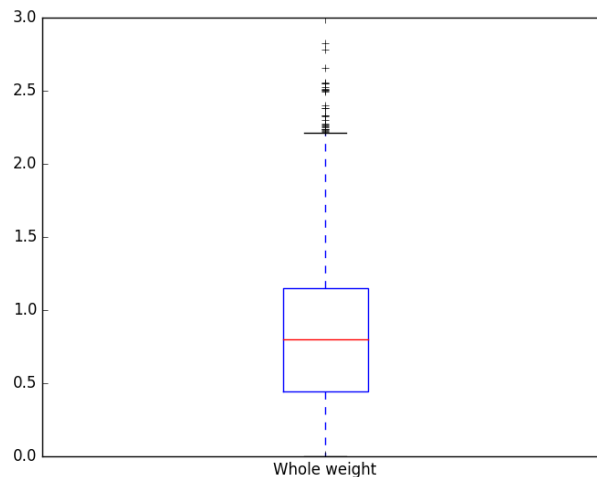
Draw the density plot for 'Shell weight':

```
In [12]: abalone['Shell weight'].plot(kind='density')  
plt.title('Abalone - Shell Weight')  
plt.xlabel('Shell weight')  
plt.show()
```



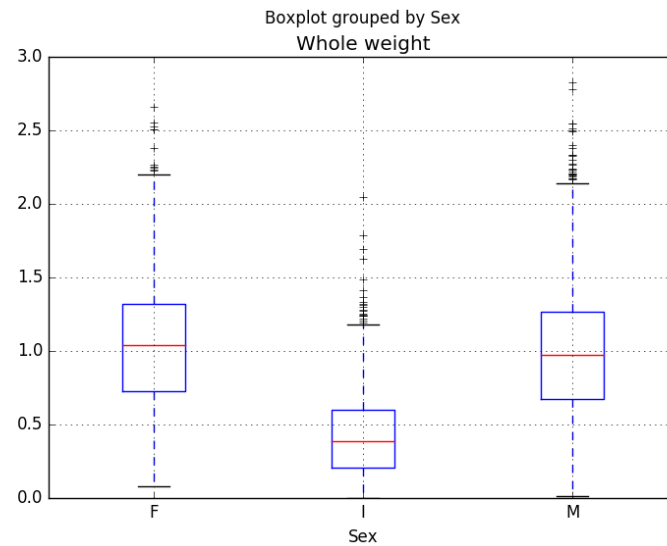
Draw the boxplot for 'Whole weight':

```
In [13]: abalone['Whole weight'].plot(kind='box')  
plt.show()
```



Draw the boxplot for 'Whole weight' per 'Sex' group:

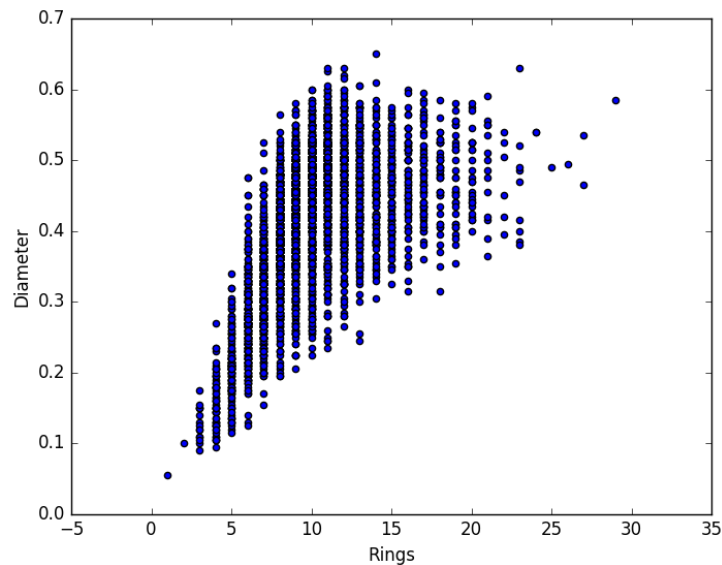
```
In [14]: abalone.dropna().boxplot(column='Whole weight',by='Sex')  
plt.show()
```



This [link](#) may help you to understand the box plot.

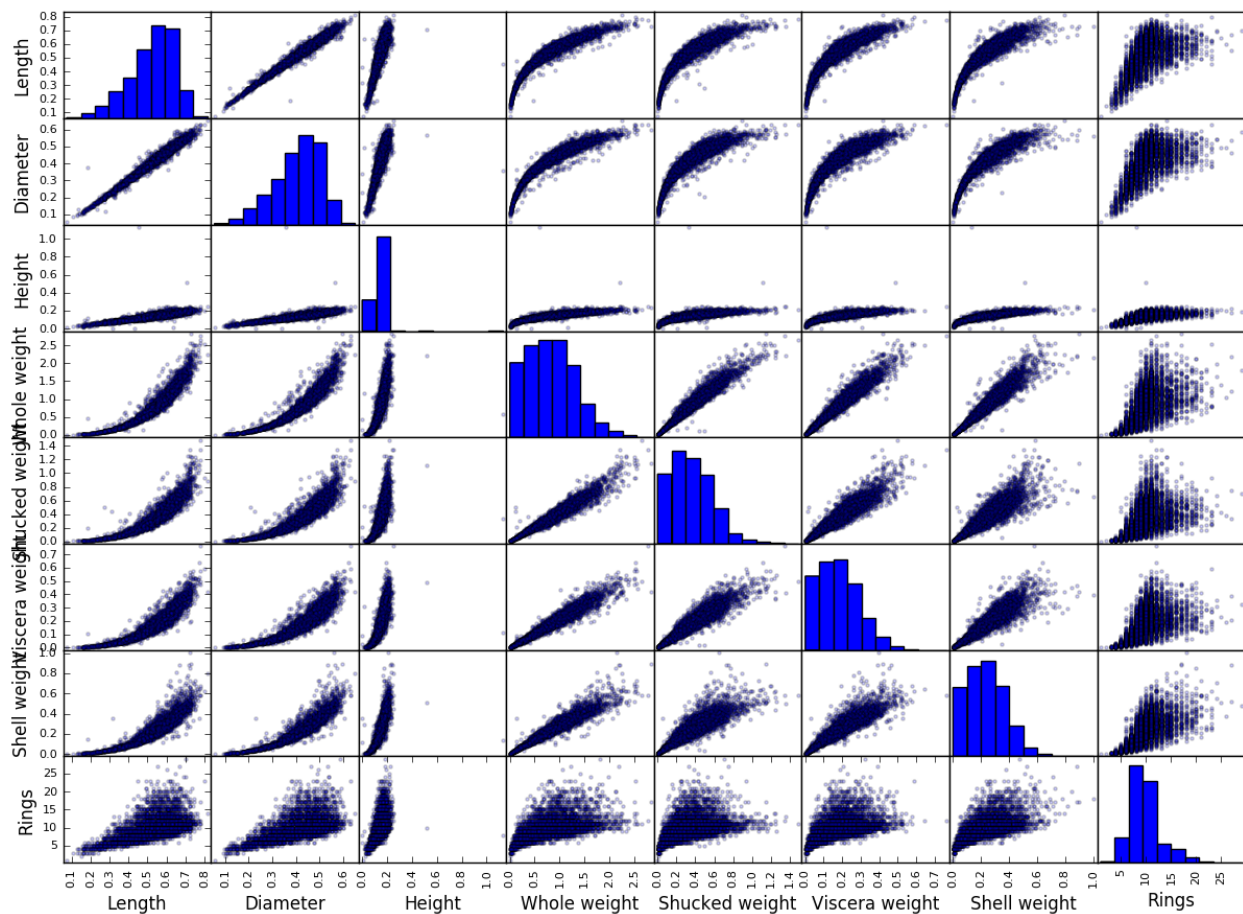
Draw the scatter plot between 'Rings' and 'Diameter':

```
In [15]: abalone.plot(kind='scatter', x=8, y=2)  
plt.show()
```



Draw the Scatter Matrix:

```
In [16]: from pandas.plotting import scatter_matrix  
In [17]: numeric_df = abalone.select_dtypes('number')  
         scatter_matrix(numeric_df, alpha=0.2, figsize=(16,16), diagonal='hist')  
         plt.show()
```



Advanced Exercise: Log Scale

- Draw the scatter plot for columns 'Length' and 'Whole weight' with and without log scale for x and y. Hint: Check the function definition on Pandas online API reference.

What do you observe?

- Now, repeat the same steps for columns 'Length' and 'Rings'. Does it have the same effect?

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